

LoGeR: DeepMind's 3D Reconstruction That Scales to 10,000 Frames with Hybrid Memory

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2026-03-19

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Google DeepMind's feedforward geometric reconstruction at kilometer scale

74% ATE reduction on KITTI vs prior feedforward methods	128→19K Frames: trained on 128 Generalizes to 19,000	448 ★ GitHub stars Code and checkpoints public
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March 19, 2026

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Google DeepMind's [LoGeR](#) (Long-context Geometric Reconstruction) scales dense 3D reconstruction from video to over 10,000 frames and kilometer-long sequences, all without any post-processing optimization. Current 3D reconstruction models face a hard tradeoff: bidirectional attention models like [VGGT](#) produce excellent local geometry but choke on long videos

due to quadratic compute costs, while recurrent models scale linearly but lose geometric coherence over time. LoGeR breaks this tradeoff with a [hybrid memory module](#) that combines two strategies. A parametric memory (Test-Time Training) adapts model weights on the fly to anchor the global coordinate frame and prevent scale drift. A non-parametric memory (Sliding Window Attention) keeps recent frames in full detail for precise alignment between consecutive chunks. Trained on just 128-frame sequences, LoGeR generalizes to thousands of frames at inference. On [KITTI](#), it cuts Absolute Trajectory Error by over 74% compared to prior feedforward methods, reaching an ATE of 18.65. On the VBR dataset (up to 19,000 frames), it delivers a 30.8% improvement over previous best results.

For anyone working in autonomous driving, robotics, or AR/VR, this is a practical step forward. Reconstructing a full driving route from dashcam footage, or building a persistent spatial map of an office from a walkthrough video, previously required expensive optimization pipelines. LoGeR does it in a single feedforward pass with linear-time scaling. The [code and checkpoints](#) are public.

Where the [Seoul World Model](#) generates video of real cities, LoGeR reconstructs their 3D geometry from video. Together, they represent two sides of the same shift: vision models that operate at city scale, not clip scale.

Sources:

- [LoGeR Paper \(arXiv\)](#)
- [LoGeR Project Page](#)
- [LoGeR GitHub Repository](#)
- [HuggingFace Daily Papers \(March 10, 2026\)](#)

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